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Burnout in health students and its relationship with demographic and academic characteristics: a survey-based study

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Abstract

Background Healthcare education demands the acquisition of clinical and interpersonal competencies under high academic pressure, exposing students to chronic stressors that may lead to burnout. Burnout among healthcare students and professionals has become a prevalent issue with significant academic and psychological consequences. This study aimed to examine the association of burnout levels among health sciences students with their sociodemographic and academic characteristics at a Chilean university.

Methods A total of 879 students from four programs—Medicine ($n = 275$; 31.3%), Physical Therapy ($n = 85$; 9.7%), Pharmaceutical Chemistry ($n = 334$; 38.0%), and Speech Therapy ($n = 185$; 21.0%)—participated through quota sampling. Participants completed the Maslach Burnout Inventory–Student Survey (MBI-SS) and a sociodemographic-academic questionnaire. Ethical approval was obtained from the institutional Ethics, Bioethics, and Biosafety Committee, and all participants provided informed consent. Descriptive analyses and multiple linear regressions were performed using gender, age, paid employment, course failure, institution type, and year of study as predictors.

Results Emotional Exhaustion (EE) showed the highest mean scores, followed by Lack of Personal Accomplishment (LPA) and Depersonalization (DP). Higher EE was associated with female gender and being in the fourth year. The regression models were statistically significant ($p < 0.05$), explaining 10.5% of EE, 5.8% of LPA, and 10.2% of DP variance. Greater LPA was linked to being female and enrolled in the second or third year, while higher DP was associated with younger age, male gender, paid employment, and being in the fourth year. Conversely, lower DP levels were observed among students from private, fee-paying schools and those studying Physical Therapy or Speech Therapy.

Conclusions Burnout among health sciences students showed distinct gender and program-related patterns. Female students and those in advanced years reported greater emotional exhaustion and diminished personal accomplishment, while male and younger students presented higher depersonalization. These findings underscore the need for targeted institutional strategies promoting psychological well-being and resilience, particularly in medicine and pharmaceutical programs.

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Keywords Burnout, Psychological, Students, Education, Medical, Stress, Psychological, Mental Health, Education, Higher, Chile, Depersonalisation, Emotional exhaustion, Personal accomplishment

Background

Education in the healthcare field demands the development of clinical and interpersonal skills in students, with an intensity that can subject them to chronic stressors with pernicious consequences, one of which is burnout. Burnout in students and healthcare workers is currently a widely studied phenomenon, and its relationship with numerous academic and sociodemographic factors such as the type of educational [1, 2] or work center [3, 4], the year of study [5–7], age [8], and gender [9] has been investigated.

The concept of burnout was introduced by Maslach [10] in the 1970s, defining it as a response to the poor management of chronic stressors when performing an occupation. In students, it is characterized by three factors: emotional exhaustion, which describes the feeling of being exhausted due to intense study demands; depersonalization, defined as a skeptical and detached attitude towards studying; and lack of personal accomplishment, marked by feelings of ineffectiveness [11–14].

Burnout occurs in various contexts, such as among teachers [15], public health officials [16], and university students [17]. Additionally, studies show high levels of burnout among health students from multiple countries, such as the United States [8, 18, 19], Canada [20, 21], Japan [22, 23], New Zealand [24, 25], Australia [26, 27], Portugal [28, 29], Italy [30, 31], Spain [32], Tunisia [12], and Chile [13], a trend that continues later in healthcare professionals [33, 34]. In these students, severe consequences of burnout have been documented, such as an increased risk of dropping out of studies, negative effects on ethical behavior and empathy, and even a higher risk of suicide [13, 14, 35], which is particularly relevant considering the high prevalence of suicidal ideation in this group [36].

The Maslach Burnout Inventory (MBI) is a widely used tool for measuring burnout and has been validated in various cultural and occupational contexts, showing adequate validity and reliability [11, 14].

Previous studies in health students report that burnout is associated with several factors. Regarding age, there is contradictory evidence; some report that a younger age is associated with higher burnout [37–39], while others suggest the opposite [8, 40], and others found no statistically significant relationship [17, 41–43]. As for the relationship with the year of study, it is generally reported that higher courses are associated with greater burnout [5–7, 35, 44–50], and some specifically report an increase in the fourth year [47–51]. In terms of gender differences, men show higher depersonalization and women higher

emotional exhaustion [9, 14, 45, 52–55]. Additionally, some studies have found a higher overall level of burnout in women [56] while others found it higher in men [57, 58].

Furthermore, studies have found significant differences in the level of burnout among students and health professionals depending on the type of center where they study or work, whether it is publicly or privately funded [2, 4, 59–63], although other studies have not found significant differences [1, 3]. Additionally, higher burnout has been linked to poor academic performance [55, 64–66], and although paid employment is a stressor for students, studies find inconsistent evidence about its effect [54, 67–72].

Although, the study of burnout in students has gained increasing interest in the scientific community, its psychosocial nature makes continued investigation necessary due to its interdependence with changes in occupational processes such as professional training, generational shifts, and even broader social transformations. In addition, there are still few studies on this phenomenon among health students in Chile. Moreover, few research have assessed its relationship with sociodemographic and academic factors in this population, particularly in the aftermath of crises that have affected medical education and student mental health, such as the COVID-19 pandemic [9, 15, 18]. Accordingly, this study aims to examine the association of burnout levels among health sciences students with their sociodemographic and academic characteristics at a Chilean university.

Methodology

For this quantitative study with an analytical-relational approach, an observational cross-sectional design was used, with data collection through a survey. Sampling The population studied corresponds to undergraduate university students from health sciences programs in Chile. We employed a non-probabilistic quota sampling method using the field of study as the segmentation criterion, resulting in a sample of 879 students from four health sciences programs: Medicine, Physical Therapy, Chemistry and Pharmacy, and Speech Therapy.

Sampling

The study population consisted of undergraduate university students from health-related programs in Chile. As inclusion criteria, we considered students who were regularly enrolled in one of the health programs studied. As an exclusion criterion, students who had been absent from their training for more than three months

during the previous semester were excluded, in order to ensure that burnout was influenced by training-related processes. No exclusions were made based on physical or mental health diagnoses. We used a non-probabilistic quota sampling method, using the program of study as the segmentation criterion. We estimated a minimum sample size of 135 students, assuming a 95% confidence interval, a power of 80%, and a medium effect size ($f^2 = 0.15$) for 14 predictors in a linear multiple regression analysis. A sample of 879 students was obtained, distributed across four health-related programs: Medicine, Physical Therapy, Pharmaceutical Chemistry, and Speech Therapy. The acceptance rate ranged from 34% in Physical Therapy to 81% in Pharmacy and Chemistry.

Instruments

To collect data, a Spanish translation of the Maslach Burnout Inventory (MBI) was used, licensed by Mindgarden Inc, its official provider, which has been previously employed with Chilean students [11, 14, 73, 74]. This version consists of 22 items grouped into the three dimensions of Maslach's model: Emotional Exhaustion, Depersonalization, and Personal Accomplishment [75]. The items correspond to descriptions of behaviors, thoughts, and feelings, to which students must respond according to the frequency with which they have experienced them, using a seven-point Likert scale (0 = Never; 1 = A few times a year or less; 2 = Once a month or less; 3

= A few times a month; 4 = Once a week; 5 = A few times a week; 6 = Every day). In previous research with Chilean students, this version has demonstrated a factorial structure consistent with Maslach's proposal, and the three subscales of this instrument showed adequate reliability with Cronbach's Alpha coefficients between 0.68 and 0.80 [11, 14]. Additionally, a sociodemographic and academic questionnaire was applied, in which students reported their gender, age, current program, year of study, type of educational establishment from which they graduated, subject failures, paid employment, among others.

Procedure

This research received approval from the Ethics, Bioethics, and Biosafety Committee of the Universidad de Concepción, under approval CEBB 1463–2023. The instruments were administered in person by the research team to the students before a theoretical class. Before responding, the students read and signed an informed consent form which explained: the general objective of the study, the names and institution of the research team, and the type of participation requested. Additionally, the confidentiality and voluntariness of their participation were guaranteed.

Analysis plan

Firstly, the score for each burnout factor was calculated by averaging the items to facilitate comparability, as the total scores vary among the three subscales. A descriptive analysis of the factors from both questionnaires was performed, calculating mean, standard deviation, median, interquartile range, minimum, and maximum, as well as absolute and relative frequencies in percentage as appropriate. The Shapiro-Wilk test was also conducted for the burnout scores, finding a non-normal distribution ($p < 0.001$). To evaluate the internal consistency of each subscale, Cronbach's Alpha and McDonald's Omega coefficients were calculated. Nonparametric Mann-Whitney U and Kruskal-Wallis H tests were used to compare burnout levels according to categorical sociodemographic and academic characteristics. Spearman's nonparametric Rho coefficient was used to calculate the correlation between burnout levels and age. Lastly, multiple linear regression analyses were applied to estimate the effect of sociodemographic and academic variables on burnout. A value of $p < 0.05$ was considered statistically significant. Data were analyzed using JASP version 0.18.3.0.

Results

A sample of 879 students was obtained, ranging in age from 18 to 34 years (Median = 21; Mean = 21.3; IQR = 4; SD = 2.5), of which 59.61% ($n = 524$) were women, and 38% ($n = 334$) were studying Chemistry and Pharmacy, as shown in Table 1.

Table 1 Description of the sample of students in the health area

Variable	Descriptive Statistics	n	%
Gender	Women	524	59.61%
	Men	355	40.39%
Program	Pharmacy and Chemistry	334	38.00%
	Medicine	275	31.29%
	Speech therapy	185	21.05%
	Physical Therapy	85	9.67%
Year	First	173	19.68%
	Second	186	21.16%
	Third	143	16.27%
	Fourth	193	21.96%
	Fifth	155	17.63%
	No response	29	3.30%
Graduating Institution	Private Subsidized	471	53.58%
	Municipal	254	28.90%
	Private paid	147	16.72%
	Foreign	4	0.46%
Paid Employment	No response	3	0.34%
	No	780	88.74%
	Yes	99	11.26%
Failed courses	No	515	58.86%
	Yes	360	41.14%
	No response	4	0.46%

N = 879; n = Absolute Frequency

Table 2 Descriptive statistics of burnout factors in health area students

Burnout Factors	ω (IC95%)	α (IC95%)	Me	M	SD	IQR	Shapiro-Wilk	Mín	Máx
Emotional Exhaustion	0.84 (0.83–0.86)	0.84 (0.82–0.85)	3.43	3.42	1.19	1.71	0.99***	0.14	6.00
Lack of Personal Accomplishment	0.75 (0.73–0.78)	0.75 (0.72–0.77)	1.75	1.81	0.80	1.00	0.98***	0.00	5.13
Depersonalization	0.72 (0.69–0.75)	0.71 (0.68–0.74)	1.43	1.59	1.06	1.43	0.96***	0.00	5.71

N=879. ω McDonald's omega, α Cronbach's alpha *Me* Median, *M* Mean, *SD* Standard Deviation, *IQR* Interquartile Range, *Min* Minimum, *Max* Maximum. *** p <0.001

Burnout levels

Regarding the burnout levels of the students, the scores showed adequate reliability in their three factors ($\omega > 0.7$), with a higher average score in Emotional Exhaustion ($M = 3.43$) and a lower one in Depersonalization ($M = 1.43$), out of a maximum average score of six points. The Shapiro-Wilk test rejected the normality of the distributions for the three factors ($p < 0.001$), Table 2.

Relationship of burnout with sociodemographic and academic variables

When analyzing the relationship between age and the average scores of the three burnout factors using Spearman's Rho statistics, no statistically significant relationships were found. On the other hand, it was found that the three burnout factors directly and significantly correlate with each other ($p < 0.001$).

Subsequently, burnout levels were compared according to the categorical variables of the study, finding significant differences according to gender in all three burnout factors; women had higher emotional exhaustion ($p < 0.001$) and lack of personal accomplishment ($p < 0.05$), while men exhibited more depersonalization ($p < 0.001$). Additionally, depersonalization was significantly higher among students who have paid employment ($p < 0.05$).

Differences in burnout scores were also found according to field of study ($p < 0.001$) and course year ($p < 0.001$). Dunn's post-hoc contrasts revealed significantly lower emotional exhaustion and depersonalization among Speech Therapy students ($p < 0.01$ and $p < 0.001$, respectively), and a greater lack of personal accomplishment in students of Chemistry and Pharmacy compared to the other three fields in the study ($p < 0.05$). Furthermore, higher emotional exhaustion was found in fourth-year students compared to other years ($p < 0.01$).

Predictive capacity of the studied variables on burnout levels

Finally, we sought to establish the joint predictive capacity of the studied variables on the scores of the three burnout factors measured by the MBI-HSS

Table 3 Multiple linear regression Results of the effect of the studied variables on emotional exhaustion

Predictors	B	95% CI	SE	r^2
Constant	3.82			
Age	-0.03	(-0.080-0.018)	0.03	< 0.01
Gender (Female)	0.53***	(0.370–0.698)	0.08	0.05
Paid employment Activity (Yes)	0.19	(-0.075-0.454)	0.14	< 0.01
Course Failure (Yes)	0.05	(-0.132-0.238)	0.09	< 0.01
Type of secondary school				< 0.01
Subsidized Private	-0.06	(-0.240-0.127)	0.09	
Private Paid	-0.06	(-0.328-0.209)	0.14	
Foreign	> -0.01	(-1131-1131)	0.58	
Course				0.04
Second	-0.18	(-0.432-0.075)	0.13	
Third	-0.17	(-0.458-0.121)	0.15	
Fourth	0.49**	(0.189–0.782)	0.15	
Fifth	0.02	(-0.320-0.360)	0.17	
Program				0.07
Physical Therapy	0.06	(-0.244-0.364)	0.16	
Speech therapy	-0.73***	(-0.981–0.471)	0.13	
Chemistry and Pharmacy	0.10	(-0.119-0.325)	0.11	

$R^2 = 0.120$ ***; Adjusted $R^2 = 0.105$. B = unstandardized regression coefficients, SE standard errors, r^2 = squared partial correlations. *N* = 879; * p <0.05; ** p <0.01; *** p <0.001

questionnaire, for which we developed three multiple linear regression models. In these models, we considered gender, age, field of study (with Medicine as the reference group), current course (with First Year as the reference), type of graduation institution (with Public as the reference), paid work activity (where 1 = Yes and 0 = No), and course failure (where 1 = Yes and 0 = No) as predictors. We treated each of the three burnout factors as dependent variables.

In the emotional exhaustion model, it was found that the set of predictors is statistically significantly associated with this factor, accounting for 10.5% of its variation ($p < 0.001$). Specifically, higher levels of emotional exhaustion were associated with being female ($p < 0.001$) and being in the fourth year ($p < 0.01$), explaining 4.71% and 4.28% of its variation, respectively. Moreover, studying speech therapy was associated with lower levels of emotional exhaustion compared to studying medicine ($p < 0.001$), explaining 6.60% of its variation, Table 3.

In relation to the lack of personal accomplishment, the group of predictors was statistically significantly

associated with this factor, explaining 5.8% of its variation ($p < 0.001$). Individually, a greater lack of personal accomplishment was associated with being female ($p < 0.05$) and being in the second or third year ($p < 0.05$). Additionally, higher levels of this factor were found in students of Chemistry and Pharmacy ($p < 0.001$) compared to those in Medicine. These three predictors accounted for 0.77%, 0.92%, and 4.97% of the variation, respectively, Table 4.

Regarding depersonalization, the predictors studied also demonstrated a statistically significant association with this factor, explaining 10.2% of its variation ($p < 0.001$). When individualizing the predictors, higher levels of depersonalization were associated with younger ages ($p < 0.05$), male gender ($p < 0.05$), the presence of paid work activity ($p < 0.001$), and being in the fourth year ($p < 0.001$), with these predictors explaining 0.64%, 0.53%, 2.02%, and 2.56% of the variation of the factor, respectively. Conversely, lower depersonalization was found in students graduating from private paid institutions ($p < 0.05$), and in students of physical therapy ($p < 0.01$) and speech therapy ($p < 0.001$) compared to their peers in medicine, explaining 0.83% and 7.02% of the variation of this factor, respectively, Table 5.

Discussion

This study aimed to examine the association between burnout levels and sociodemographic and academic characteristics among health sciences students at a Chilean university. Our findings indicate that burnout levels were bivariately associated with gender, field of study, academic year, and having paid employment, whereas no significant association was observed with age or the type of secondary school attended. Several of these variables remained significantly associated with burnout after adjustment for potential confounders in the multivariable regression analyses.

Despite not finding a relationship between age and burnout levels using Spearman's Rho, when developing the regression models, a younger age did prove to be predictive of higher depersonalization. Although the bivariate results align with other studies that have not found a relationship between these variables [41, 42], they also coincide with the limitation of having used bivariate statistics. Moreover, given the inconsistency of previous studies that show age appears not to be related to burnout levels [17, 43], or to be related to higher or lower burnout [8, 37–40], this could indicate that directly evaluating age may not be the best approach, since burnout is associated with other factors that could be measured independently, such as coping mechanisms, emotional management, level of maturity, etc.

Regarding the relationship between burnout levels and paid employment, our findings associated higher levels of depersonalization with the presence of paid work.

Table 4 Multiple linear regression results of the effect of the studied variables on lack of personal accomplishment

Predictors	B	95% CI	SE	r ²
Constant	1.82			
Age	-0.01	(-0.046–0.021)	0.02	< 0.01
Gender (Female)	0.15*	(0.034–0.257)	0.06	< 0.01
Paid Employment Activity (Yes)	0.14	(-0.041–0.319)	0.09	< 0.01
Course Failure (Yes)	-0.07	(-0.192–0.059)	0.06	< 0.01
Type of secondary school				< 0.01
Subsidized Private	-0.10	(-0.220–0.029)	0.06	
Private Paid	-0.13	(-0.308–0.057)	0.09	
Foreign	0.42	(-0.352–1.185)	0.39	
Course				< 0.01
Second	0.23*	(0.055–0.399)	0.09	
Third	0.21*	(0.009–0.402)	0.10	
Fourth	0.18	(-0.023–0.380)	0.10	
Fifth	0.14	(-0.091–0.372)	0.12	
Program				0.05
Physical Therapy	0.06	(-0.150–0.262)	0.11	
Speech therapy	-0.15	(-0.323–0.024)	0.09	
Chemistry and Pharmacy	0.33***	(0.174–0.477)	0.08	

R² = 0.272***; Adjusted R² = 0.058. B = unstandardized regression coefficients, SE standard errors, r² = squared partial correlations. N = 879; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 5 Multiple linear regression results of the effect of the studied variables on depersonalization

Predictors	B	95% CI	SE	r ²
Constant	2.83			
Age	-0.05*	(-0.095–0.008)	0.02	< 0.01
Gender (Female)	-0.16*	(-0.303–0.010)	0.08	< 0.01
Paid Employment Activity (Yes)	0.50***	(0.259–0.730)	0.12	0.02
Course Failure (Yes)	-0.14	(-0.306–0.024)	0.08	< 0.01
Type of secondary school				< 0.01
Subsidized Private	-0.14	(-0.302–0.025)	0.08	
Private Paid	-0.25*	(-0.493–0.014)	0.12	
Foreign	0.61	(-0.403–1.613)	0.51	
Course				0.03
Second	0.14	(-0.085–0.367)	0.12	
Third	0.24	(-0.014–0.502)	0.13	
Fourth	0.56***	(0.297–0.826)	0.14	
Fifth	0.26	(-0.048–0.559)	0.15	
Program				0.07
Physical Therapy	-0.36**	(-0.630–0.089)	0.14	
Speech therapy	-0.73***	(-0.960–0.506)	0.12	
Chemistry and Pharmacy	< 0.01	(-0.195–0.201)	0.10	

R² = 0.117***; Adjusted R² = 0.102. B = unstandardized regression coefficients; SE = standard errors; r² = squared partial correlations. N = 879; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

However, although some studies have linked employment with higher levels of stress [67, 69], they did not directly correlate it with burnout. Other studies have also linked more burnout with the presence [68, 72] or absence [70] of paid employment, or simply found no significant differences between burnout and work [54, 71], so there is

no consensus in the literature. This could indicate that engaging in paid employment means less time, energy, and more stress, which can affect students' academic performance [67–69] and lead to a decline in their physical, emotional, and mental state, along with the development of a skeptical and detached attitude towards studies [68]. However, paid work can also have positive effects on academic performance by reinforcing discipline and the sense of academic responsibility, as well as improving the perception of a lack of time [71]. Therefore, depending on the individual, paid employment can be either a stressor or a protective factor against burnout. It would be interesting to investigate why this duality occurs and how to promote the conversion of paid employment from a stressor to a protective factor in working students.

On the other hand, the relationship between burnout levels and the year of study, we found that being in the fourth year was associated with the highest levels of emotional exhaustion and depersonalization. On the other hand, the greatest lack of personal accomplishment was associated with being in the second or third year. The results are consistent with the available evidence in the literature, as various studies link higher burnout in students who are in higher academic years [5–7, 35, 44–50], particularly highlighting the fourth year in programs that last between 5 and 7 years [47–51]. This suggests a non-linear relationship between the year of study and burnout, but with critical points during the educational trajectory that expose students to higher burnout. This could be related to the fact that the fourth year marks the entry into clinical practice, which represents a significant change in the course of health-related careers, often leading to increased stress and anxiety [76–78]. Considering this, we can suggest that it would be appropriate for institutions to strengthen the approach to the profession in pre-clinical courses, such as early practices, and to better prevent or prepare health area students for the emotional exhaustion and depersonalization they will face when they reach the clinical practice courses.

Gender comparisons found that men experience more depersonalization, compared to women who suffer more from emotional exhaustion. This aligns with previous literature that has found similar results [9, 14, 45, 52–55]. This helps to better understand burnout as male and female students face the same stressors, and the variation likely relates to how men and women tend to approach problems [55]. In practice, university psychologists, when treating students, should be aware that the way of coping with stress differs between genders. Since depersonalization tends to lead to more evasion [79], men may experience negative consequences because evasion can occur through mechanisms such as alcoholism, which is more common in men [80]. When working with women, it should be recognized that emotional exhaustion may

be associated with the onset of other mental health issues such as depression [37, 81], implying that different approaches are needed for each gender.

In relation to the field of study, lower levels of emotional exhaustion were found in speech therapy students, a greater lack of personal accomplishment in chemistry and pharmacy students, and less depersonalization in physical therapy and speech therapy students, all compared to medicine. The results obtained are expected according to previous literature that has found differences between fields, where medicine has shown higher levels of burnout, although it differs from other studies that found that burnout affects all types of health students equally [34–36]. These results help to understand that different fields present different stressors that lead to the various dimensions of burnout, requiring particular attention in medicine which shows greater depersonalization and emotional exhaustion compared to other health fields. In practice, this allows us to know, according to the field of study, which burnout factor to pay more attention to and provide targeted support so that the student can cope with these stressors.

A lower level of depersonalization was found among students who graduated from private, fee-paying schools, which represent the wealthiest groups in the population and whose educational system tends to be more successful in student formation. These results are consistent with other studies that found significant differences in the level of burnout among students and healthcare professionals according to the type of institution they belong to, whether public or private [2, 4, 59–63]. However, they differ from other studies where no significant differences were found [1, 3]. Our results suggest that students who graduated from private, fee-paying schools probably possess more tools to cope with burnout provided by their educational institution, which leads to lower depersonalization among these students. In practice, this provides evidence to pay special attention to students from other types of schools to understand how they cope with burnout and provide them with tools to compensate for their initial disadvantages, which their previous educational institution may not have provided.

No relationship was found between course failure and the three factors of burnout. These results differ from previous studies where a relationship between burnout and poor academic performance was found [55, 64–66]. This helps to understand that academic failure does not necessarily lead to burnout.

Conclusion

To summarize, the results of this study indicate that factors such as being a woman, being in the fourth year of study, and studying medicine are associated with higher levels of emotional exhaustion. It was found that being a

woman, being in the second or third year, and studying Pharmacy are associated with a greater lack of personal accomplishment. Likewise, higher levels of depersonalization were associated with factors such as being a man, being younger, having a paid job, and being in the fourth year of study. In contrast, lower levels of depersonalization were associated with students who graduated from private, fee-paying schools and students studying Physical Therapy and Speech Therapy.

On the one hand, these results contribute to expanding knowledge about burnout among health sciences students in Chile. Specifically, in institutions such as the one studied—characterized by standardized and highly selective admission processes—students are trained in programs lasting five years, such as Speech Therapy and Physical Therapy; five and a half years, such as Chemistry and Pharmacy; or even seven years, as in Medicine. In these programs, training during the initial years takes place primarily in university-based settings, with a gradual introduction to clinical environments beginning in the third or fourth year. On the other hand, these findings may help guide the development and implementation of strategies aimed at preventing burnout among health sciences students, with particular attention to groups at greater risk, considering the multiple factors associated with its different components.

Limitations and future lines of research

One of the limitations of this study is the use of a self-administered questionnaire. This type of instrument only allows for screening and identifying subjects with high scores on the Burnout scale, but it does not allow for establishing a diagnosis of Burnout. To diagnose it, as a suggestion for future research, a clinical evaluation could be carried out, for example, having the subjects assessed by a psychiatrist or psychologist. Another limitation concerns the way the study sample was selected, since a non-probabilistic quota sampling method was used. As the subjects were not selected randomly, there could be some discrepancies between the study results and what actually happens in the general population. Therefore, it is suggested that future research consider selecting study participants using probabilistic sampling.

An additional possible limitation is that the study was conducted in only one university in Chile and exclusively with students from the health sciences field. The educational conditions of this group of students are different from those of other fields of knowledge. Therefore, it is recommended that future research expand the study to more universities and to programs from other disciplines in order to have a broader understanding of the behavior of Burnout in university students as a whole, and to provide a basis for the development of new and/or improved

intervention strategies for the prevention of burnout in students.

Abbreviations

MBI	Maslach Burnout Inventory
MBI-SS	Maslach Burnout Inventory–Student Survey
EE	Emotional Exhaustion
LPA	Lack of Personal Accomplishment
DP	Depersonalization
ANID	Agencia Nacional de Investigación y Desarrollo (National Agency for Research and Development, Chile)
FONDECYT	Fondo Nacional de Desarrollo Científico y Tecnológico (National Fund for Scientific and Technological Development)
SD	Standard Deviation
IQR	Interquartile Range
CI95%	95% Confidence Interval
ω	McDonald's Omega
α	Cronbach's Alpha

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Authors' contributions

Conceptualization: CPV, RGL. Methodology: CPV, APM, MIR-O. Investigation: APM, DVR-P, AHP-Z, POB. Data curation: APM, MIR-O. Formal analysis: MIR-O, APM. Visualization: POB, AHP-Z. Writing original draft: APM, MIR-O. Writing review & editing: CPV, RGL, BS, JFHQ. Supervision: CPV, RGL. Project administration: RGL, CPV. All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ethical and privacy considerations but are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval was obtained from the institutional Ethics, Bioethics, and Biosafety Committee of Universidad de Concepción (CEBB No. 1463–2023), and all participants provided written informed consent. The study was conducted in accordance with the Declaration of Helsinki, the Universal Declaration of Human Rights, the Ethical Norms of the Pan American Health Organization for Research with Human Subjects, the Political Constitution of the Republic of Chile, Law No. 20.120 "On Scientific Research in Human Beings, Their Genome and the Prohibition of Human Cloning", and Law No. 19.628 "On the Protection of Private Life". It also adhered to the Suggestions for Writing an Informed Consent in Studies with Human Participants formulated by the FONDECYT/CONICYT Bioethics Advisory Committee and adopted by the National Agency for Research and Development (ANID) of the Ministry of Science, Technology, Knowledge and Innovation of the Government of Chile.

Consent for publication

Participants were made aware of the purpose of the research and how their data would be used, ensuring their confidentiality and anonymity throughout the process. All participants provided informed consent.

Competing interests

The authors declare no competing interests.

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